

EXAMPLE 11.12.3. On the great-circle route connecting JFK and LAX, find the geographic coordinates of the waypoint that is 1,000 nautical miles from JFK.

Solution. From Example 11.12.1 or 11.12.2, the geographic coordinates are

$$\text{JFK: } (0.709185, -1.287762) \quad \text{LAX: } (0.592539, -2.066470)$$

Using these values in Eq. (11.12.41) gives

$$\Theta = 2 \sin^{-1} \left[\sqrt{\sin^2 \left(\frac{\Phi_d - \Phi}{2} \right) + \cos(\Phi) \cos(\Phi_d) \sin^2 \left(\frac{\Psi_d - \Psi}{2} \right)} \right] = 0.623585$$

$$\Theta_{wp} = d_{wp} / R_E = 0.290888$$

$$A = \sin(\Theta - \Theta_{wp}) / \sin(\Theta) = 0.559283$$

$$B = \sin(\Theta_{wp}) / \sin(\Theta) = 0.491144$$

$$x = A \cos(\Phi) \cos(\Psi) + B \cos(\Phi_d) \cos(\Psi_d) = -0.075245$$

$$y = A \cos(\Phi) \sin(\Psi) + B \cos(\Phi_d) \sin(\Psi_d) = -0.765932$$

$$z = A \sin(\Phi) + B \sin(\Phi_d) = 0.638503$$

$$\left. \begin{aligned} \Phi_{wp} &= \text{atan2} \left(z, \sqrt{x^2 + y^2} \right) = 0.692551 \\ \Psi_{wp} &= \text{atan2}(y, x) = -1.668721 \end{aligned} \right\} \underline{(39^\circ 40' 49'' \text{ N}, 95^\circ 36' 39'' \text{ W})}$$

11.13. Problems

- 11.1. Write a computer subroutine to evaluate the quaternion product of two arbitrary quaternions. Use the general relation given by Eq. (11.6.3).
- 11.2. Write a computer subroutine to evaluate the components of the Euler-Rodrigues quaternion from known values of the traditional aircraft Euler angles. Use the positive sign in Eq. (11.7.8). Test your subroutine against the result that was obtained in Example 11.7.1.
- 11.3. Write a computer subroutine to determine the traditional aircraft Euler angles from known values of the components of the Euler-Rodrigues quaternion. Use Eq. (11.7.9) and remember that the arctangent function in most programming languages returns a value in the range $-\pi$ to π , whereas the heading angle is traditionally defined in the range 0 to 2π . Test your subroutine against the result that was obtained in Example 11.7.1.
- 11.4. Write a computer subroutine that uses the Euler-Rodrigues quaternion and the subroutine written for problem 11.1 to transform the components of an arbitrary vector from body-fixed coordinates to Earth-fixed coordinates. Test your subroutine against the result that was obtained in Example 11.7.1.